

AKSHAR AI

SUMMER INTERNSHIP REPORT

WEEK 1 | 25 MAY 2026 – 29 MAY 2026

Project Title:

Chest X-Ray Pneumonia Detection and Analysis System Using Deep Learning

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1. Introduction

Pneumonia is a critical respiratory condition caused by bacterial, viral, or fungal infections that inflame the alveoli of the lungs. It remains one of the leading causes of mortality globally, particularly among children below the age of five and elderly populations. Timely and accurate diagnosis is, therefore, of paramount clinical importance. Chest X-ray (CXR) imaging is the most widely employed frontline diagnostic modality due to its accessibility, speed, and cost-effectiveness. However, manual interpretation of CXR images demands sustained expert attention and is susceptible to inter-observer variability, especially in high-volume clinical environments.

Convolutional Neural Networks (CNNs) and Transfer Learning architectures have demonstrated remarkable proficiency in medical image classification tasks, offering a reliable computational alternative to manual radiological analysis. This internship project at Akshar AI Ltd focuses on designing and deploying a comprehensive deep learning pipeline that processes raw chest X-ray images and classifies them as either Normal or Pneumonia, while also integrating Explainable AI (Grad-CAM) overlays to ensure clinical transparency.

This report covers the activities completed during Week 1 of the internship, spanning from 25 May 2026 (Monday) to 29 May 2026 (Friday). The primary focus of this week was on two foundational tasks: Task 1 — Dataset Loading and Understanding, and Task 2 — Exploratory Data Analysis (EDA). The outputs of these tasks were subsequently integrated into a React-based web application for structured visual presentation, and all project assets were pushed to a Git repository for version control and collaborative access.

2. Weekly Objectives

The following technical objectives were defined for Week 1 of the internship project:

- Acquire the chest X-ray dataset from Kaggle and establish a structured working environment in the project directory.
- Verify dataset integrity by checking for corrupted, missing, or invalid image files across all class folders.
- Perform statistical and spatial analysis to understand class distribution, image resolution profiles, and pixel intensity characteristics.
- Identify structural artifacts (e.g., orientation labels, surgical hardware) embedded within the X-ray images that could cause shortcut learning.
- Develop a React-based frontend application to display the analysis results in a structured and interactive format.
- Commit and push all project files to a remote Git repository for version control.

Day	Date	Objective
1	25 May 2026 (Mon)	Project orientation, dataset overview, tech stack setup
2	26 May 2026 (Tue)	Kaggle dataset extraction, initial EDA, data analysis technique study
3	27 May 2026 (Wed)	Task 1: Directory mapping, file counting, metadata extraction, dataset validation
4	28 May 2026 (Thu)	Task 2: Visual grids, class distribution plots, resolution & pixel analysis, artifact audit
5	29 May 2026 (Fri)	React frontend integration, EDA result display, Git push

Table 1: Week 1 Day-wise Objective Summary

3. Day-wise Work Summary

Day 1 – Monday, 25 May 2026

Project Orientation and System Setup

The first day was devoted to project onboarding and environment configuration. The project was formally introduced as the Chest X-Ray Pneumonia Detection and Analysis System — an end-to-end deep learning pipeline designed to perform binary classification of chest radiographs as either Normal or Pneumonia.

The assigned technology stack was reviewed and confirmed: Python (TensorFlow/Keras with OpenCV and PIL for image operations), React.js for frontend development, and Kaggle as the primary data source. The development environment was configured, including the installation of required Python libraries and Node.js packages. The System Requirement Proposal document was reviewed to gain a thorough understanding of the project scope, task breakdown, and expected deliverables. The overall eleven-task implementation roadmap was studied, with a focus on the Week 1 milestones: Task 1 (Dataset Loading and Understanding) and Task 2 (Exploratory Data Analysis).

Day 2 – Tuesday, 26 May 2026

Dataset Extraction and Preliminary Exploration

On the second day, the chest X-ray dataset was successfully downloaded from Kaggle using the Kaggle API. The dataset, originally structured across three directories — train, test, and val — contains grayscale JPEG images labelled under two categories: NORMAL and PNEUMONIA. The canonical directory structure was reviewed, and folder paths were confirmed to be accessible for subsequent processing tasks.

A preliminary examination of the dataset was conducted to gain an initial understanding of its composition. Basic file counts were noted, and a selection of X-ray samples were visually inspected to identify gross differences between the two classes. Supplementary study of standard data analysis techniques was conducted through a YouTube tutorial to ensure that the EDA phase would be approached with methodological rigor.

Day 3 – Wednesday, 27 May 2026

Task 1: Dataset Loading and Understanding

Day 3 was entirely focused on executing Task 1 as defined in the System Requirement Proposal. The objective was to programmatically load, map, validate, and profile the Kaggle chest X-ray dataset.

- **Directory Mapping:** Python libraries — os, pathlib, and glob — were employed to navigate the dataset file system and verify the existence and accessibility of the train, test, and val directories, along with their respective NORMAL and PNEUMONIA subdirectories.
- **File Enumeration:** The total count of image files in each class folder was recorded to assess initial class balance. A significant imbalance was observed between the PNEUMONIA and NORMAL classes within the training partition, with the pneumonia class containing substantially more samples.
- **Dataset Validation:** Systematic validation checks were performed to identify corrupted, missing, unsupported, or zero-byte image files. Validation was conducted by attempting to open each file using PIL's Image.verify() method, flagging any entries that raised exceptions.
- **Metadata Extraction:** A representative subset of images was loaded using OpenCV and PIL. Key metadata attributes were extracted: image width, height, channel depth, colour mode, and file format. The analysis revealed that the majority of images were grayscale (single-channel), with dimensions varying considerably across samples.

Partition	NORMAL	PNEUMONIA	Total
Training Set	1,341	3,875	5,216
Validation Set	8	8	16
Test Set	234	390	624
Total	1,583	4,273	5,856

Table 2: Dataset Distribution Across Partitions and Classes

Day 4 – Thursday, 28 May 2026

Task 2: Exploratory Data Analysis

Day 4 was dedicated to performing a comprehensive visual and statistical analysis of the chest X-ray dataset as outlined in Task 2 of the System Requirement Proposal. The following analyses were conducted:

- **Dynamic Grid Visualization:** Matplotlib was used to generate automated image grids displaying multiple NORMAL and PNEUMONIA chest X-ray samples in alternating rows. This enabled a side-by-side visual comparison of lung morphology across both classes, making it possible to observe key radiological differences such as consolidation patterns, opacity regions, and alveolar fluid in pneumonia-affected images.
- **Class Distribution Analysis:** Seaborn bar plots were generated to illustrate the ratio of NORMAL to PNEUMONIA images across all data splits. The charts confirmed a notable class imbalance in the training set, with pneumonia images outnumbering normal samples by approximately 2.9:1. This finding is critical for selecting appropriate loss functions and sampling strategies during model training.
- **Resolution Profile Analysis:** Scatter plots mapping image height against image width were produced to characterize the distribution of native image resolutions. The analysis revealed considerable dimensional heterogeneity, reinforcing the need for standardized interpolation to a fixed 224×224 input size during the preprocessing phase.
- **Pixel Intensity Analysis:** Grayscale intensity histograms were extracted and plotted to contrast the pixel density profiles of NORMAL versus PNEUMONIA images. Normal lung images generally exhibited higher pixel uniformity and lower average intensity values across the lung fields, whereas pneumonia-affected images displayed broad high-intensity regions corresponding to fluid opacities and consolidation zones.
- **Structural Artifact Audit:** A targeted inspection was performed to detect non-biological annotations embedded within the X-ray images, including orientation markers (e.g., 'R', 'L', 'PORTABLE') and visible surgical hardware such as pacemaker leads and chest drains. These artefacts were documented as potential sources of shortcut learning, where a model may learn to associate the presence of such markers with a particular class label rather than genuine pathological features.

Day 5 – Friday, 29 May 2026

React Frontend Integration and Git Version Control

On the final day of the week, the analysis outputs from Task 1 and Task 2 were consolidated and integrated into a React-based web application. The objective was to present the dataset exploration results in a structured, interactive, and visually coherent format that would be accessible to stakeholders and team members without requiring them to execute Python scripts directly.

The application was structured into clearly delineated sections: a dataset overview panel displaying total image counts and partition summaries, a class distribution section featuring bar charts, a sample image gallery with labeled thumbnails, an image metadata table, and a pixel intensity histogram viewer. The backend-generated results were consumed by the React frontend through structured data files, ensuring seamless rendering of charts and statistical summaries.

Upon completing the frontend implementation, all project files — including the Python analysis scripts, EDA output files, and the React application source code — were committed to the Git repository. Proper commit messages were authored to document the progression of work. The repository was pushed to the remote host, ensuring the availability of a versioned backup and facilitating future collaborative development with team members.

4. Methodology

The methodology adopted for Week 1 followed a structured pipeline approach aligned with the project's System Requirement Proposal. The workflow progressed through three logical phases:

Phase 1: Data Acquisition and Environment Setup

The dataset was sourced from Kaggle's Chest X-Ray Images (Pneumonia) repository and downloaded programmatically using the Kaggle API. Directory structures were established and validated using Python's `os`, `pathlib`, and `glob` libraries to ensure consistent file system navigation across all subsequent tasks.

Phase 2: Statistical and Visual Data Profiling

File enumeration and validation scripts were developed to count image files per class and check for corrupted or missing entries. Metadata extraction routines used PIL and OpenCV to collect image dimensions, channel depths, and format strings. Statistical visualization was performed using Matplotlib and Seaborn to produce class distribution charts, resolution scatter plots, and pixel intensity histograms. Artifact detection was conducted through manual visual sampling supported by programmatic file inspection.

Phase 3: Frontend Presentation and Version Control

EDA outputs were structured as JSON/CSV data files and consumed by a React.js frontend. Components were developed for each analysis section, with Recharts or Chart.js integration for interactive graph rendering. Upon completion, all code and assets were committed and pushed to Git using descriptive commit messages following standard naming conventions.

5. Tools and Technologies Used

Category	Tool / Technology	Purpose
Language	Python 3.x	Primary scripting language for data loading, EDA, and validation
Image Processing	OpenCV, PIL (Pillow)	Loading images, extracting metadata, pixel-level operations
File System	os, pathlib, glob	Directory traversal, file enumeration, path management
Visualization	Matplotlib	Image grids, histogram plots, resolution scatter plots
Visualization	Seaborn	Statistical bar charts for class distribution analysis
Frontend	React.js	Interactive web application for displaying EDA results
Data Source	Kaggle API	Dataset acquisition and download automation
Version Control	Git / GitHub	Source code versioning, backup, and collaboration
IDE / Notebook	VS Code / Jupyter	Code editing, script execution, and inline visualisation

Table 3: Tools and Technologies Used in Week 1

6. Results and Observations

The following key findings were obtained from the Task 1 and Task 2 analyses during Week 1:

6.1 Dataset Composition

The Kaggle Chest X-Ray dataset comprised a total of 5,856 images distributed across three partitions. The training set exhibited a pronounced class imbalance, with the PNEUMONIA class containing approximately 2.89 times more images than the NORMAL class. The validation set was notably small (16 images total), which necessitates careful handling during the model evaluation phase to prevent misleading validation metrics.

6.2 Image Quality and Metadata

The dataset contained predominantly JPEG-format, single-channel grayscale images. Native image resolutions varied widely, with widths ranging from approximately 300 pixels to over 2,000 pixels. No zero-byte or severely corrupted image files were identified during the validation phase, indicating a relatively clean dataset suitable for direct preprocessing without significant data cleaning overhead.

6.3 Visual Class Differences

Grid visualisations confirmed clearly observable differences between NORMAL and PNEUMONIA X-rays. Normal lung images exhibited well-defined, uniformly dark lung fields with clear costophrenic angles. Pneumonia-affected images displayed irregular white patches, diffuse opacities, and consolidation regions, most prominently in the lower lung zones, consistent with classical radiological presentation.

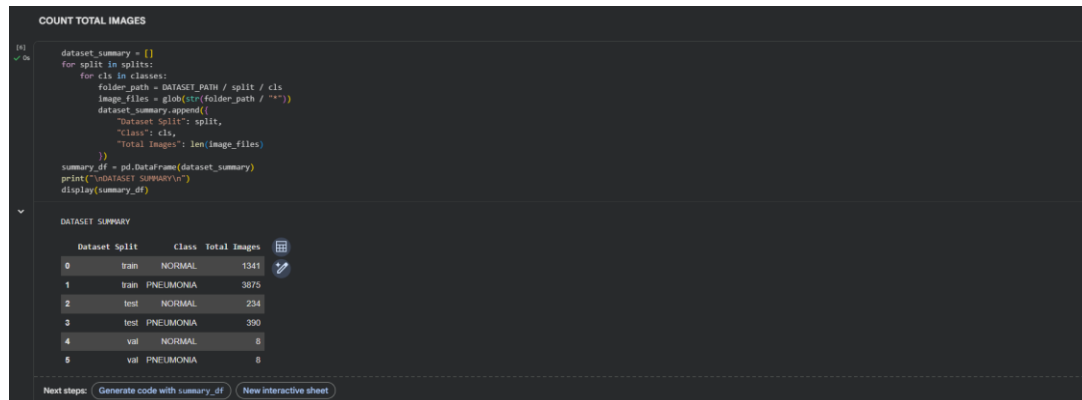
6.4 Pixel Intensity Profiles

Characteristic	NORMAL Images	PNEUMONIA Images
Dominant Pixel Range	0 – 80 (dark lung fields)	80 – 200 (opacities present)
Histogram Shape	Left-skewed, low intensity	Broader, right-shifted peaks
Consolidation Evidence	Absent	Present (bright patches)

Table 4: Comparative Pixel Intensity Characteristics

6.5 Artifact Observations

A subset of images was found to contain embedded text markers such as 'R', 'L', and 'PORTABLE', typically printed at the image corners. Additionally, certain images included metallic hardware artefacts including pacemaker leads and chest drain tubes. These non-pathological patterns were documented for potential masking or preprocessing treatment in the data augmentation phase to mitigate shortcut learning risks.



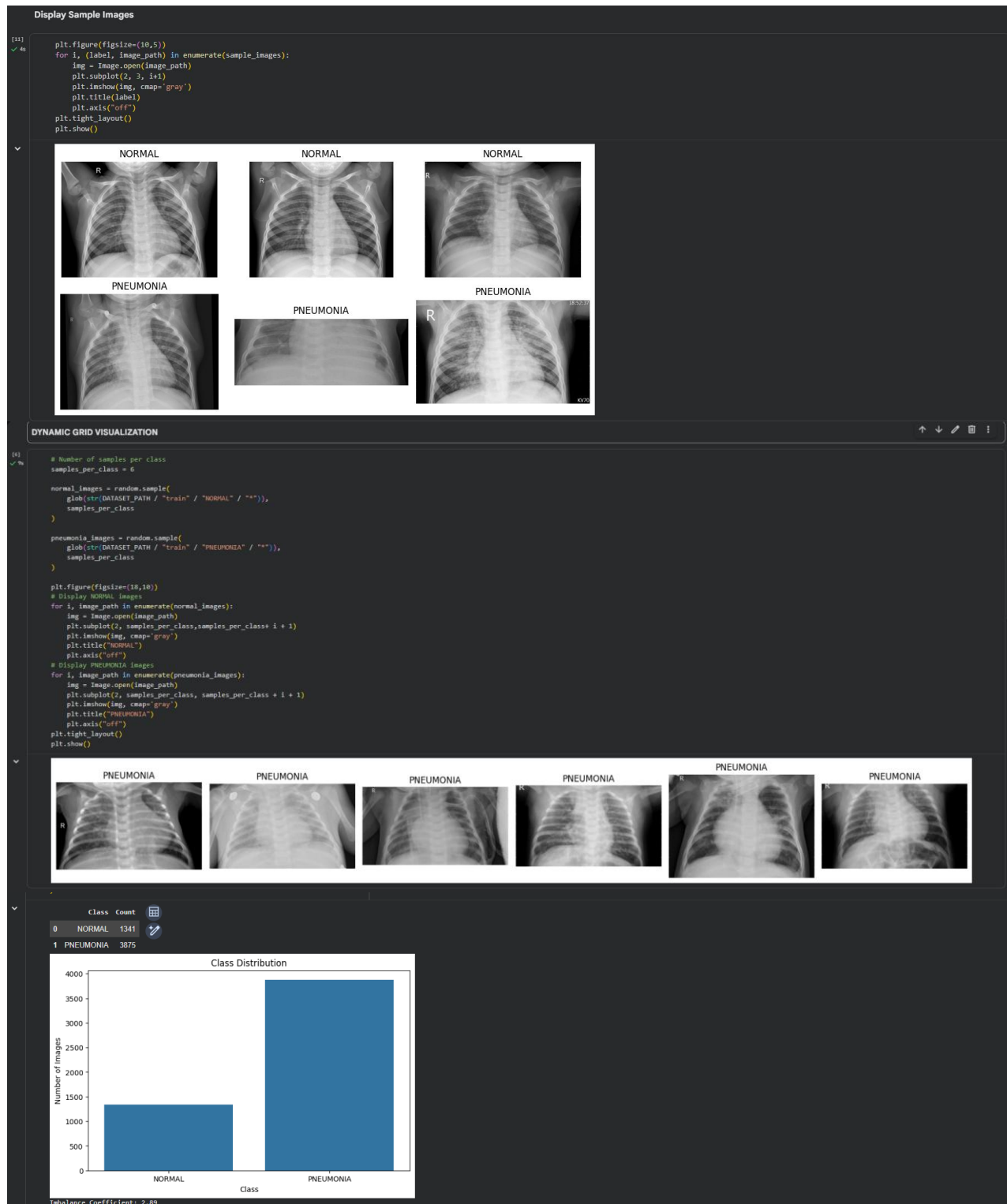
COUNT TOTAL IMAGES

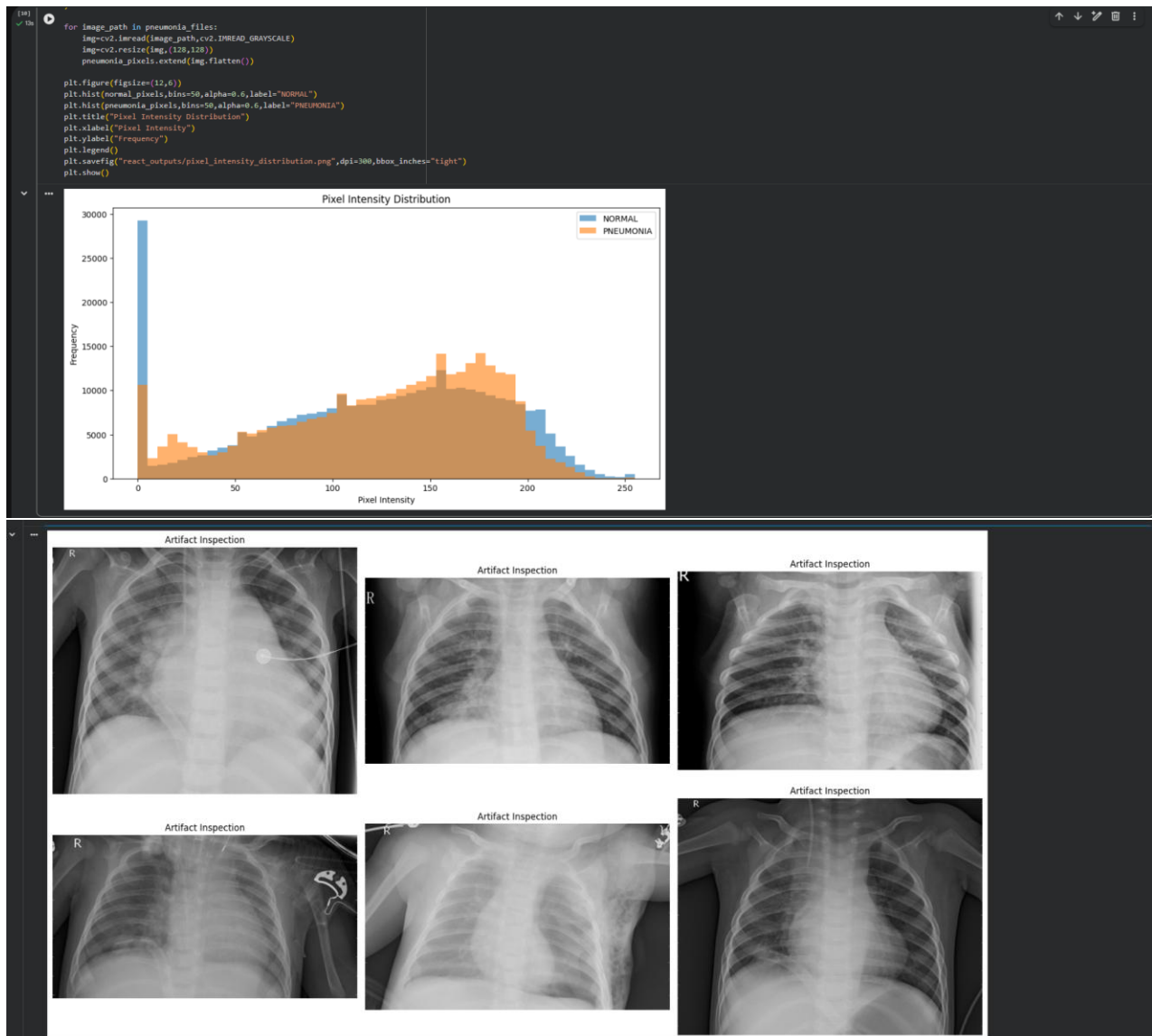
```
(1) dataset_summary = []
(2) for split in splits:
(3)     for cls in classes:
(4)         folder_path = DATASET_PATH / split / cls
(5)         image_files = glob.glob(folder_path / "**")
(6)         dataset_summary.append(
(7)             "Dataset Split": split,
(8)             "Class": cls,
(9)             "Total Images": len(image_files)
(10)         )
(11) summary_df = pd.DataFrame(dataset_summary)
(12) print("\nDataset Summary:")
(13) display(summary_df)
```

DATASET SUMMARY

	Dataset Split	Class	Total Images
0	train	NORMAL	1341
1	train	PNEUMONIA	3875
2	test	NORMAL	234
3	test	PNEUMONIA	390
4	val	NORMAL	8
5	val	PNEUMONIA	8

Next steps: [Generate code with summary_df](#) [New interactive sheet](#)





7. Challenges Faced

- **Class Imbalance:** The significant imbalance between NORMAL (1,341) and PNEUMONIA (3,875) images in the training set presents a risk of biased model learning. Addressing this imbalance through oversampling, class weighting, or augmentation strategies will be critical in subsequent weeks.
- **Resolution Heterogeneity:** The wide variation in native image dimensions required careful interpolation strategy selection. Uniform downsampling to 224×224 pixels was identified as necessary, but the choice of interpolation method (bilinear vs. bicubic) needed evaluation to minimize information loss.
- **Artifact Identification:** Distinguishing between clinically relevant opacities and non-biological artifacts in radiographs required careful manual inspection. Automated detection of such artifacts remains a non-trivial preprocessing challenge.
- **React Integration:** Rendering large sets of EDA visualisations (charts, image grids, histograms) efficiently in the React frontend required thoughtful state management and component structuring to avoid performance bottlenecks during page load.

- **Small Validation Set:** The Kaggle dataset's validation split contained only 16 images (8 per class), which is statistically insufficient for reliable generalization monitoring. An alternative validation strategy, such as k-fold cross-validation or a custom re-split of the training data, will need to be considered.

8. Learning Outcomes

Week 1 of the internship yielded the following technical and professional learning outcomes:

- Developed hands-on proficiency in programmatic dataset acquisition using the Kaggle API and in structured file system navigation using Python's `os`, `pathlib`, and `glob` modules.
- Gained practical experience in medical image analysis, including pixel-level interpretation of chest X-ray features and the identification of radiologically significant patterns such as consolidation and opacity.
- Strengthened understanding of class imbalance as a data-level challenge in medical AI and its downstream implications for model training and evaluation strategies.
- Acquired experience with EDA visualisation libraries (Matplotlib and Seaborn), including the construction of multi-panel image grids, distribution plots, and statistical histograms.
- Developed understanding of shortcut learning risks arising from non-biological artifacts in medical imaging datasets, and the importance of pre-training data audits.
- Enhanced skills in React.js frontend development, particularly in structuring data-driven components and integrating backend analytical outputs into interactive web interfaces.
- Reinforced Git version control best practices, including meaningful commit messages, structured repository organisation, and remote push workflows for team-based collaboration.

10. Conclusion

Week 1 of the Summer Internship at Akshar AI Ltd established a solid technical and analytical foundation for the Chest X-Ray Pneumonia Detection and Analysis System. Through the systematic execution of Task 1 (Dataset Loading and Understanding) and Task 2 (Exploratory Data Analysis), a comprehensive profile of the Kaggle chest X-ray dataset was constructed, capturing key attributes including class distribution, image resolution variance, pixel intensity characteristics, and the presence of structural artifacts.

The EDA revealed a statistically significant class imbalance in the training partition and a wide range of native image resolutions, both of which must be addressed during the preprocessing phase in Week 2. The pixel intensity analysis confirmed observable radiological differences between NORMAL and PNEUMONIA images, providing a data-driven basis for the model's classification task.

The integration of all analytical outputs into a React-based web application and the successful push to the Git repository ensured that the Week 1 deliverables were not only technically sound but also well-documented, accessible, and version-controlled. The internship experience thus far has reinforced core competencies in medical image data engineering, EDA methodology, frontend development, and professional collaborative workflows. The groundwork laid in Week 1 positions the project for a structured and efficient transition into the model development and training phases in the weeks ahead.

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